

- [3] *Software Inspections*—Tom Gilb and Dorothy Graham (Addison Wesley)
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- [5] *Software Engineering Productivity*—C. Stevenson (Chapman & Hall)
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- [8] *The Art of Software Testing* – Myers (John Wiley & Sons)
- [9] *STQE Volume 4, Issue 2, March/April 2002* (The Software Testing and Quality Engineering Magazine)
- [10] *A Slice-by-Slice Guide to Total Quality Management*—John Gilbert (Affiliated East-West Press Pvt. Ltd)

SOFTWARE CONFIGURATION MANAGEMENT

7.1 Overview

"Change Management" is one of the fundamental activities of software engineering. This is because the business environments for which software applications are developed, themselves are very dynamic. Especially the rate of change in today's global economy and Internet days is beyond imagination. As a result of this, the business requirements and expectations from software applications keep changing. In fact, in the world of software project management, they say that change is the only reality. Changes to the requirements drive the design, and design changes affect the code. Testing then uncovers problems that result in further changes, sometimes even to the original requirements. Though the change process may sound simple in concept, it is quite complex in real-life large projects involving multi-teams at multi-locations. The number of communication channels in a team of the size n are: $[n(n-1)/2]$. The Number of co-ordination problems also increases exponentially with the project size. This is because the number of channels will rise exponentially as the team size increases. For even modest-sized projects, the number of people involved and the change volume generally require a formal change management system. That is the reason why there is a need for Software Configuration Management (SCM as popularly known). A key objective of the software process is to have change activity converge until the final product is stable enough to ship. Given this, it should be noted that management of ALL changes is important. Among change management activities, code control is typically the main focus but it is important to note that it is NOT the only activity under SCM. Control of requirements and design changes is also of critical importance. These functions are often handled as enhancements to code management system.

7.2 Configuration Management: Why and What

Before we start the discussion, here are some basic definitions pertinent to the topic of this chapter:

CONFIGURATION: *the arrangement of a computer system or components as defined by the number, nature, and interconnections of its constituent parts.*

CONFIGURATION MANGEMENT: *a discipline applying technical and administrative direction and surveillance to: identify and document the functional and physical characteristics of a configuration item, and control changes to those characteristics, report change processing and implementation status, and verify compliance with specified requirements.*

To know why configuration management, consider the following aspects of software:

- Software is Easy to Copy!
- Changes Are frequent
 - Different Users/Customers have different Requirements
 - Changes get clarified/known at a later date
 - Business Environment, too, changes
 - Technology Changes
 - Personnel whims!
- Software Projects: Personnel Turnover
 - Developers leave
 - Users change
 - Customer Single Point Contact Changes

All this further adds to the problems...already faced by the Project
Main Classes of Problems Software Projects are:

- Double maintenance Problems
- Shared Data
- Simultaneous Updates
- Missing/Unknown Version Problem

For example, life of a typical project manager/project leader is full of anecdotes like the ones below:

- Listing seems OK: program does not work
- Works in Mumbai but not Chennai
- We had customized for this client. How do we install the upgrade now?
- I'd fixed this bug last month; How did it re-appear?
- I haven't changed the program. Why is it blowing up?
- Which is the latest source? I need to put a patch
- In the last month, the user asked for this change and she doesn't want it

...the program has been working on?

All this only means that change is inevitable when computer software is built. And change increases the level of confusion among software engineers working on the project. When change are not analyzed and documented before implementing then, confusion will result. Before making the changes, they need to be informed to those who need to know them (the testers, the end users etc). Thus, changes need to be controlled in a manner that will improve quality of the deliverable and reduce error.

.....The art of coordinating software development to minimize... confusion is called "configuration management". Configuration management is the art of identifying, organizing, and controlling modifications to the software being built by a programming team. The goal is to maximize productivity by minimizing mistakes.....

Roger Pressman [1] describes Software Configuration Management (SCM) to be an umbrella activity that is applied throughout the software process. Because change can occur at any time, SCM activities are developed to (1) identify change, (2) control change, (3) ensure that change is being properly implemented, and (4) report changes to others who may have to do with it.

It is important to make a clear distinction between software support and software configuration management. Support is a set of software engineering activities that occur after software has been delivered to the customer and put into operation. Software configuration management is a set of tracking and control activities that begin when a software engineering project begins and terminate only when the software is taken out of operation. Let us provide an IEEE definition of Software Configuration Management:

A discipline applying technical and administrative direction and surveillance to: identify and document the functional and physical characteristics of a configuration item, control changes to those characteristics, record and report change processing and implementation status, and verify compliance with specified requirements.

This definition of Configuration Management includes six points:

1. It is a **Management** discipline
2. It **Identifies** the proposed or implemented (actual) configuration of a system
3. This is done at discrete points in time
4. Systematically **records and traces** changes to all system components (conceptual and physical)
5. Provides tools for **controlling** changes; and finally
6. Allows everything happening with (and to) the system, throughout the entire life-cycle of the system, to be **verified** via auditing and reporting tools

All this is for purposes of assuring: integrity, accountability, visibility, reproducibility, project co-ordination and traceability and formal control of system/product evolution. In the context of this book, Configuration Management is a process used for efficiently developing and maintaining software. This is accomplished by improving: accountability, reproducibility, traceability and co-ordination.

"Configuration" is a list of parts and their relative arrangement. This includes the number, nature and interconnections of all constituent parts. To understand Configuration Management in the context of software systems, let us discuss what Software is:

- o Structured information with hierarchical, logical and functional properties
- o Created as text but maintained in several, and frequently parallel, representations during the life cycle of the product
- o Machine 'processable' in its most advanced state
- o Maintained in various forms, and with different tool, during development use, maintenance and operations
- o Composed of parts which are themselves software
- o Created via the use of tools, which are themselves also software

7.2.1 Nomenclature

Since the role of configuration management is to control the development of the system element as they are built and then combined/integrated into a full system, it is important to establish a common system terminology before proceeding in the discussion on the topic of Software Configuration Management. This is depicted in the figure 1.

- **System** The package of all the software that meet's the user's requirements

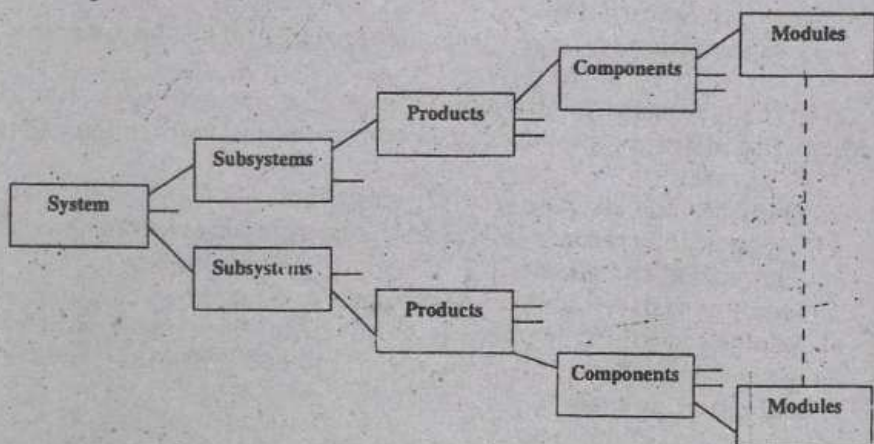


Figure 1: Software Product Nomenclature

- **Subsystem** Large systems can have many subsystems, such as communication, display and processing.
- **Product** subsystems typically contain many products. For example, an operating system might contain a control program, compilers, utilities and so forth.
- **Components** At the next level, a control program could be made up of such components of the sub-system.
- **Module** At the lowest level, components consist of a number of modules. Modules typically implement functions that are relatively small and self-contained.

7.3 Software Configuration Management Activities

The discipline of software configuration management has four major activities: **Identification, Control, Status Accounting and Auditing**. Here's a brief explanation to each of these activities. This will be followed by detailed treatment of each in subsequent sections.

- **Identification:** refers to the general structure of the product being produced and how the items (modules, classes etc) are to be identified
- **Control:** refers to the methods to be implemented for the management and technical control of myriad configuration items identified. (What is a "configuration item" will be defined later in this section.)
- **Status Accounting:** refers to reporting to all concerned—management, client and technical—information describing items and their status.
- **Auditing:** refers to activities, which are designed to assure the correct functioning of the SCM system, as defined.

This particular breakdown has been accepted as an industry standard. Firstly it was accepted by the IEEE as a document. Later this standard was accepted by the American National Standards Institute. The full history of IEEE is outside the scope of this book. The next sections will discuss each of the aspects of Configuration Management activities briefly.

7.3.1 Software Configuration Identification

Configuration Identification is described as: An element of configuration management, consisting of selecting the configuration items for a system and recording their functional and physical characteristics in technical documentation.

This is a formal definition as taken from part of the IEEE standard. A few points to ponder follow. The process of selecting configuration

items is not quite as simple as it may seem. It is not merely a process of identifying that the system has (say) 50, or 500 files. This may be sufficient for small systems consisting of up to (say) 100 files. This begins to more complex when the system is large (several thousands files) and/or the interrelationships between the files are complex. As a hypothetical example, consider the following scenario:

A system which is composed of (say) four sub-systems, each of which is composed of several modules, each module being composed of several code units, which in turn also have several "include" files, all these contained in individual files. Also, there is a common set of 'library' routines, which everyone uses. In addition to this, we must maintain:

- Documentation
- The specifications and designs
- The test designs, data and reports
- The change and problem reports
- The 'make' files which direct and document the way the system is constructed (sometimes called 'build' files etc.)

This is a very small example of how the level of complexity rises, even for as small a system as in this example, due to both the quantities of individual items to be created and maintained and the complexities of their interrelationships. Thus, it is not just a question of simply 'selecting' the configuration items', but rather of deciding upon a general method for the process of selecting. Thus, the rules governing the process of software configuration identification (SCI) are:

- SCI defines the 'granularity' of the SCM (the concept of a generalized method for identification).
- SCI defines what is needed to be 'seen' i.e. the visibility to all those who need to know what is happening on the project.
- SCI ensures that the identification scheme selected reflects the structure of the product, the project and the organization.
- The identification process must always be coupled with a parallel process of labeling the item with a distinct and unique label.
- SCI is a critical project management task.

Let us understand what a "work product" is. Work products are tangible artifacts that are produced during the project. This includes models, reports, diagrams, plans, code and other documents, which are direct 'stepping stones' to the final deliverable. They have a specific purpose in the engagements and describe specific content using predefined semantics and syntax. Work products are produced as a result of performing one or more tasks. Some tasks produce less tangible outputs (e.g. pass or fail, trained staff) and are called 'task outcomes'.

Thus, we can say that a work product is an artifact that gets created in various phases of a (software) project. Some of the work products may undergo changes during the tenure of a project and so they need to be kept either under document control or configuration control. Thus, those work products that are likely to undergo change should be identified as "configuration item".

Thus, *Software Configuration Item* can be defined as work product or information that is created as part of the software engineering process. Given below are a few examples of work products that can be identified as configuration items. Note that there can be slightly different practices varying from organization to organization. For example, the last three work products mentioned below may be kept under document control.

- Requirements specifications
- Design Specifications
- Test Plans and Test Cases
- Source code
- Software Tools
- Software Configuration Management Plans
- Software Verification and Validation Plans
- User Manuals

Often there is confusion about what should be put under 'Document Control' v/s what should be under 'Configuration Control'. Any work product that is likely to undergo change during the tenure of the project (until the point of shipment), and forms as a deliverable to the customer and has to do with the way the customer is going to operate the given system, should be under configuration control. A work product other than this can be just under document control. This is purely author's view based on experience of practices seen adapted in the industry. The discussion of scope of a document control system is not within the scope of this chapter. Any standard reference on ISO 9001:2001 can be referred to understand the expectations from "document control".

7.3.1.1 Important Identification Principles

Now that we have gained a basic understanding of the concepts behind configuration identification, let us examine some principles that govern the process of identification of software items.

LABELS

Software items are labeled at discrete points in time. In a computerized environment, a date/time stamp is quite natural. From this we can say that *each item's label is related to labels of predecessors and siblings.* This

relationship must be made obvious. It should have some logical hierarchical numbering scheme. For instance, an item labeled '2.3' clearly is later than something labeled '2.1' and earlier than '2.5'. This number is used here for illustrative purpose only. The label will usually consist of two parts; a name and a version number. The number is the time-related part, as it changes over time with each instance of the particular part being revised. The name must remain constant over time for this item.

BASELINING

The IEEE defines a baseline as:

- (1) *A specification or product that has been formally reviewed and agreed upon, that thereafter serves as the basis for further development, and that can be changed only through formal change control procedures.*
- (2) *A document or set of such documents formally designated and fixed at a specific time during the life cycle of a configuration item.*

Baseline through Analogy

Here is a wonderful analogy provided by Roger Pressman [1]

Consider the door to the kitchen in a large restaurant. One door is marked OUT and the other is marked IN. The doors have stops that allow them to be opened only in the appropriate directions

If a waiter picks up an order in the kitchen, places it on a tray and then realized he has selected the wrong dish, he may change to the correct dish quickly and informally *before* he leaves the kitchen through the door marked OUT

If, however, he leaves the kitchen, gives the dish to the customer and then is informed of his error, he *must* follow a procedure established by the restaurant: (1) look at the order placed to determine if an error has indeed occurred (2) apologize profusely, (3) return to the kitchen through the door marked IN, (4) explain the problem to the Chef, get the dish changed as per client specification stated on the order and so on.....

Now with this analogy in mind, let us continue to discuss some more principles:

Partitioning a baseline is always subjective. So far we have discussed individual items. As shown in Figure 1, items can be aggregate of items. The topmost aggregate item is the product or system. Anytime this item (i.e. the system) is 'released', this is called a *baseline*.

Create some partition and maintain it. Whatever partition is decided upon, remember that once you have created a partitioning of the

baseline, redoing it can be an enormous job. This is an early decision in the software project and all concerned must be consulted to make a 'correct' decision right from the beginning.

Baselines live forever! Once a baseline is created-defined-released for the first time, it is known as "input baseline". It (the input baseline) can never change! Subsequently, however, if a change has to be made, then it is a new baseline.

Revisions v/s Versions The difference between these two is a frequent source of confusion. This discussion refers to the numeric part of the label discussed earlier. The common industry wisdom seems to be this: 'revisions' refer to items that are 'internal' (to the developing organization) while the 'version' refers to released product out of the organization.

Visibility and Traceability. 'Visibility' means permitting the software to be seen by anyone who is allowed to see it: permitting the management to 'see' what is happening and permitting the management to be seen on a project; that is, permitting them to really manage. 'Traceability' is the ability to link individual events and parts to each other in time backward as well as forward.

7.3.1.2 Configuration Identification Numbering

For the purposes of 'identification of items', a system of configuration identification numbers is both most common and most convenient. In most cases (except for very large and complex software systems) the numbering system should be kept simple. Typically, a numbering system for identifying the revisions of each item accompanied by a mnemonic is sufficient.

The most common method is the following:

- The version/revision number consists of two parts, separated by a dot
- The first part (that is, the part to the right of the dot) denotes the current revision from the last baseline.
- The second part (to the left of the dot) denotes the number of the last baseline
- Generally, the right-hand number will be fixed with each baseline

7.3.2 Software Configuration Control

We have stated that Software Configuration Management is a discipline intended to facilitate the management of projects. This means that in order to control software, the process of change must be controlled. The IEEE formal definition of the process is as follows:

Configuration Control: An element of configuration management, consisting of the evaluation, co-ordination, approval or disapproval, and

implementation of changes to configuration items after formal establishment of their configuration identification.

While setting up the configuration management scheme for the organization of the project, the first thing will be that levels of control must be defined. Remember! *What you identified you must control!*

Software Configuration Control solves four of the most common problems facing software developers: the *double maintenance problem*, the *shared data problem*, the *simultaneous update problem* and the *missing/unknown version problem*. These problems are discussed in the following sections.

Double Maintenance problem is characterized by the following:

- Multiple copies of same software in use
- Fix in one—SHOULD FIX IN OTHERS
- Example
 - Same set of common routines in two systems
 - Same system installed at multiple sites
 - * forgot to inform
 - * sites detect bug at same time and "fix" the same bug differently!
- What is really required:
 - The Bug gets fixed in all copies
 - Fix same bug in identical manner in all copies

Shared Data Problem means:

- Changes made in one program interfere with proper functioning of other program
 - Example: subroutines, classes, DBMS definitions, header files, third party routines (DLLs/OCXs)
- Need
 - Control on modifications
 - Good communications

Simultaneous Update means:

- One module being worked on by more than one developer
 - Changes made by one developer disappear
- Need
 - Better division into modules
 - Ensure no simultaneous working

Missing/Unknown Version Problem means:

- Consciously decide which version to keep, which to destroy
- Use systematic method to identify versions and changes across versions
- Use consistent back-up procedures

7.3.3 Software Configuration Status Accounting

Basically, the main issue of Software Configuration Management is acquisition and maintenance of all information concerning a project's status and that of its parts. This status information must then be available for reporting to various levels of authority (i.e. anyone with the need to know!) Thus one needs to ask these questions: **WHAT HAPPENED? WHEN DID IT HAPPEN? WHAT ARE THE REASONS? WHO AUTHORIZED THE CHANGE? WHO PERFORMED THE CHANGE? WHAT ITEMS WERE AFFECTED?**

Time is always a critical part of status information. For each configuration item, regardless of whether the item is an individual item or an aggregate, a separate 'logical account' is maintained and transactions are recorded. Typically, a register/journal is required (either manually maintained or in form of a computer-based format) for recording these transactions. Moreover, transactions are not the only information which is recorded. In addition to the transactions, actual programming information is recorded upon the configuration item's archive. Reports are produced from the transaction log, from the archives, or from a combination of them. The section below describes the minimal list of reports that are part of the activity called "Configuration Status Accounting".

7.3.3.1 Configuration Status Accounting Reports

There are three reports, which must be called the 'most basic of reports', and must always be available. These are:

1. Transaction log
2. Change log
3. Item 'delta' report

Some other typically common reports are:

4. Resource usage
7. Stock status (status of all configuration items)
8. Changes in progress
9. Deviations agreed upon
10. Audit trail of all activities
11. File revision history
12. Traceability

The critical points to be aware of when attempting to define reporting requirements are:

- How much formality does the customer require?
- Who is audience for each report?
- Are project standards sub-ordinate to corporate standards?

7.3.3.2 Change Control Board

For discussion in the sections ahead, an important term needs to be described: CHANGE CONTROL BOARD (CCB) also called CONFIGURATION CONTROL BOARD (CCB) – *a group of people responsible for evaluating and approving/disapproving proposed changes to configuration items, and for implementation of approved changes.*

Typically, CCB members are:

- Project Managers
- User Representative
- Quality Controller
- Configuration Controller (appointed by the project manager)

In the following section, Change Log, Delta Report, Stock Status Report, Transaction Log are discussed.

7.3.3.3 Change Log

This log is implemented as an informal/formal notebook (the degree of formality would depend on the size and scope of the software project). However, preference in normal circumstances is to implement this log as computerized register/journal. In any case, the implementation method is always dependent upon the nature of the computing resourcing available with software project organization.

The log should contain all information relevant to, and regarding, requested changes in the system or any part thereof. The minimal distribution frequency of this report is monthly or event-driven (say a major release etc). Regardless of when the previous distribution of this report happened, this report will be (additionally) distributed before every formal review. The following minimum information is expected in the change log:

- Change number
- The change number will be assigned by the CCB (Change Control Board as described above) or the configuration controller
- Change request status type:
 - o Request Opened
 - o Pre-process evaluation
 - o Authorized for implementation
 - o In implementation
 - o In pre-release testing
 - o Released-originator notified of change
 - o Processing suspended
 - o Change request denied

- Change Request Originator details (name, designation, contact numbers, organization submitting the CR – Change Request etc)
- Impact Analysis brief– i.e. software elements or documents affected by this change request (Configuration Item number along with its version number)
- Change Request Origination date (the date the CR is first submitted)
- Change Description (A brief description of the submitted change)
- Details of the person who will carry out the Change Request when approved (the name, contact details his/her organization etc)
- Impact Analysis Details – the interfaces and impacts of the submitted change; including as applicable:
 - o The impact on other systems
 - o Configuration items
 - o Other contractors
 - o System resources
 - o Training etc
 - o Documentation
 - o Hardware
- Follow-up action required to implement this change request (when approved by CCB)

7.3.3.4 Delta Report

This report is generated to summarize progress of the development and to compare this progress with status presented in the previous report. Also this report displays the differences between version n of a document/module and version $n + 1$. The report should include:

- Dates (the dates/reporting period of the report)
- Narrative (a summary of work performed during the reporting period including progress of each task or unit of work as of the end of the reporting period.) For example:
Line i /module k change on dd.mm.yy
Change resulted from change no. [____]
Staff Hours invested for change evaluation
Staff Hours invested for change implementation
- Format:
The description should be by tasks, and the breakdown of tasks and/or units of work should conform to the WBS—work breakdown structure as shown in the example below. (For basic grounding in WBS concepts and other minimal project management concepts, reader can browse through reference [2] quoted at the end of this chapter.)

Additional WBS No. affectedAdditional Configuration Items affected

7.3.3.5 Stock Status Report

This report should summarize the status of each and all system changes which are in a defined state of 'OPEN'; and should indicate the following items:

- **Inventory:**
A list of all the incomplete changes in the system
- **Description:**
For each change, a brief description of the change should be included
- **Status:**
Information related to change status
- **Analyst/programmer/software engineer:**
The name, contact details and organization of the individual responsible for the change
- **Completion date:** the date for completing the change

7.3.3.6 Transaction Log

The transaction log is implemented as an informal notebook/register/journal, when lack of computing resources in a software project do not allow for a better (i.e. computerized) option. Preferably, this log is implemented as a computerized register. In any case, the method of implementation would depend on the computing resources available with the software project organization.

The log contains comments of, and concerning activities, of the project with respect to Configuration Management. This report should clearly show the effect and relationships resulting from each and every event (Change Request, Change Notification etc.), which occurred during the course of the software project. At the end of this section (Table 1) we've provided a format for *Change Request*.

The purpose of Transaction Log is to provide visibility to all those concerned with the project and it should have the following minimum entries:

- **Transaction number** (a consecutive number for each and every entry on the log)

- Date and Time stamp (date and time of then entry and of the event)
- Originator (name of the person submitting the entry in the transaction log)
- Software element(s) or system part(s) affected
- Description of the activity (a brief description of the activity and conditions including relevant information)
- Change in Progress Report
- Participants: (the names of the persons involved in this activity)
- Impact (the impact of this activity, if any, on other planned activities of the project)
- Follow-up if required

Table 1: Change Request Form

Change Request

(To be filled in by person initiating request)

Initiated by: Date:

Project ID:

Type of Change:

Customer requirement

Review comments to be incorporated

Defects Identified

Bug Fix

Others (specify change)

Change description: (attach additional sheets, if necessary)

(To be filled in by the Project Leader/Person making the change)

Change request ID (filled in by the project leader):

Impact of change :

Sl. no.	CI s to be changed	Responsibility	Effort required
---------	--------------------	----------------	-----------------

Planned start: Planned end:

Resources required:

Approver's Name:

Signature:

Reason for rejection:

Change completed on:

Verified by:

7.3.4 Configuration Auditing

A software configuration audit should periodically be performed to ensure that the SCM (Software Configuration Management) practices and procedures are rigorously followed. This audit can be performed directly by the specially appointed SCM staff or by some independent assurance function.

Every major baseline/release *must* be audited. For in-house systems, an informal audit is usually sufficient. Formal audit does not mean something that is unfriendly or antagonistic; the audit function is much too important for it to be relegated to someone who is not capable of performing in a manner conducive to healthy relationships. Periodic reviews *must* be held to determine progress. Of all project management functions, change control always is the most important thing to audit! Table 2 below presents a checklist for configuration audit.

Table 2: A Checklist for Configuration Audit

Sr. No	Description	Yes	No	NA
<i>Configuration Items (CIs)</i>				
1.	Are the Configuration Items identified for the project?			
2.	Is the responsible person for each CI identified?			
3.	Are the CIs uniquely identified? Can changes to each CI be traced?			
<i>Configuration Control Board (CCB)</i>				
4.	Does the CCB exist for the project?			
5.	Are all the activities specified in the plan carried out by CCB?			
6.	Does the Configuration Controller (CC) exist for the project?			
7.	Are the activities of CC carried out as per plan?			
<i>Baselining</i>				
8.	Have the baselines identified?			
9.	Are the CIs comprising each baseline identified?			
10.	Has an input baseline established?			
11.	Are the baselines established as per plan?			
12.	Are the activities during base-lining carried out as per plan?			
13.	Is the tool to establish baseline and store and recover archive versions of configuration items identified and used?			
14.	Is the tool identified is used to establish baseline and store and recover archive versions of configuration items?			
15.	Is the integrity between the software baselines maintained?			
<i>Configuration Management Library System (CMLS)</i>				
16.	Is the Configuration Management Library System established?			
17.	Does the CMLS support generation of reports?			
18.	Are the access rights on each CI for different individuals established in CMLS as per the plan?			

Version Control

19. Is the Version Control procedure followed as defined in the plan?
20. Is the version numbering scheme and incremental scheme defined in the plan implemented?
21. Are the change requests handled as per the procedure defined in the plan?

Backup & Recovery

22. Has the Backup been taken as per plan?
23. Do the Backups exist at appropriate location as per plan?
24. Are the CIs recovered from archive at any time?
25. Is the recovery taken place according to the procedure defined in the plan?

Configuration Status Reporting

26. Are the status reports generated as per plan?
27. Are the status reports sent to the appropriate persons as per plan?
28. Are the status reports accessible to all affected individuals?
29. Do the reports give the proper status of the system?

Configuration Audits

30. Is the configuration audits taken place as per schedule?
31. Have the non-conformances observed during the previous audit been closed?

Synchronization (if applicable)

32. Is the code handled at multiple locations?
 33. Is the synchronization of changes at the defined frequency as per plan?
 34. Is the synchronization activity logged?
-

In this context, the term 'audit trail' is very important – let us understand what it means and why it should be a major concern. The idea is to provide as full a trace ('footprint') as possible. The intention is to provide 'project visibility'. All Software Configuration Management activities must leave some sort of residue upon the audit trail file. At the very least, all modifications to any archive (sometimes called a log file) are certainly to be reported. Audit trail information must contain four basic types of fields:

- Id of the group/individual member of the team issuing the command
- The command that was issued
- Which archive(s) have been processed
- What actions have occurred as a result of the command

Generally, configuration auditing is best performed by an external auditor i.e. someone external to the project team. Why—Because a very high degree of objectivity is required when auditing a critical management function. Figure 2 shows the audit process.

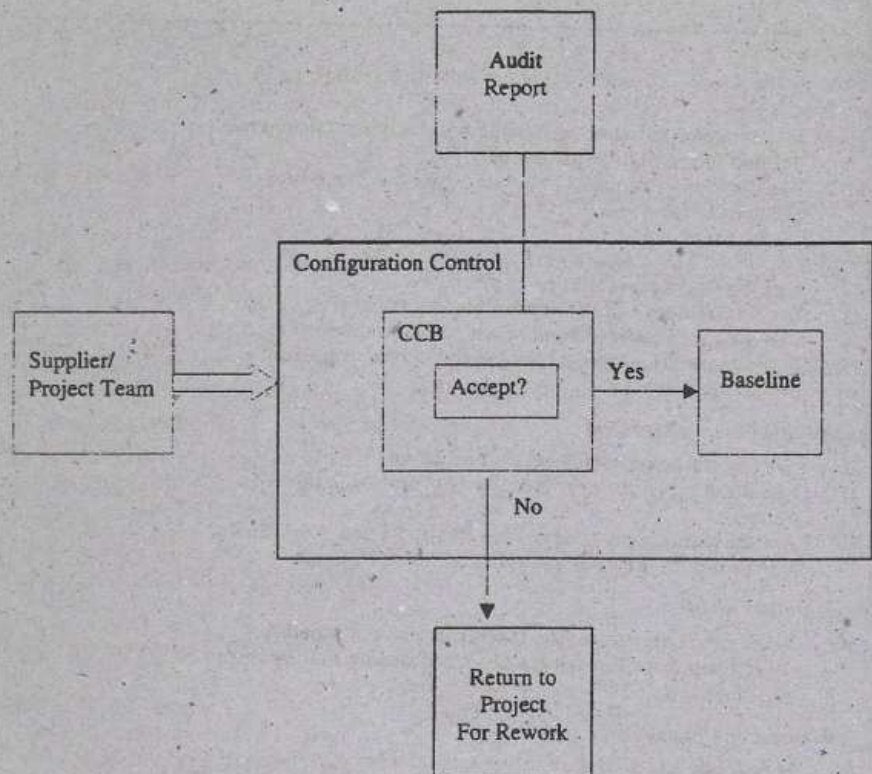


Figure 2: The Configuration Audit Process

7.3.4.1 Control over Suppliers and Subcontractors

It should be noted that the items or subsystems, which are developed by sub-contractor or supplied by vendor, must always be controlled and audited for Configuration Management. The main contractor of the project, or the customer, must be the one to determine their CM procedures.

The determination of sub-contractor Configuration Management requirements should be governed by a consistent concept of the needs of the agency issuing the contract. These requirements must be a part of the statement of work (SoW) issued to the subcontractor as part of the contract signed or work order issued. Sometime, there are 'cultural' differences in terms of maturity between the supplier/subcontractor and customer organization. The SCM requirements must be tailored to the product being developed but as a flow-down or result of those requirements governing the whole system. All SCM requirements for the subcontractor must be a result of co-ordination with the engineering functions, logistics and product/program management. Under no circumstances can the demand for a computerized SCM system be

relinquished vis-à-vis the subcontractor. This is particularly true for software libraries. All issues and documentation governing the configuration identification and control of interfaces must be very well established and frequently audited.

7.3.4.2 Guidelines for Configuration Audit

With the discussion above in background, here're some general guidelines for SCM audits:

- SCM Audits are periodically needed to ensure the integrity of the software baselines.
- The SCM requirements should be adapted to the size, types and quantities of configuration items being developed.
- An audit is performed before every major baseline change or major release and non-conformances if any are tracked to closure and reported to all stake holders.
- The configuration audit team consists of qualified technical people who are not involved in the specific tasks being audited.
- The audit verifies that changes to the baseline are implemented as intended.
- The auditing function is an integral part of the SCM system.
- SCM auditing is continuous; with increased frequency and depth as development progresses.
- A documented project SCM plan is used as the basis for all SCM audits.

SCM audits are required prior to key checkpoints in each development phase. A phase review process then ensures that the proper actions are taken as follows:

1. *Requirements*—At the requirements phase review, SCM releases the software requirements and operational concept documents and places them under change control.
2. *Functional*—At the specification phase review, SCM releases the software specification document(s) and places it under change control.
3. *High-level design*—At the preliminary design phase review, SCM reviews the preliminary design documents, assigns configuration items as defined for the software components, and updates the configuration item index. SCM releases the preliminary design document, the software test plan, and the user documents, and places them under change control.
4. *Detailed Design*—At the detailed design phase review, SCM reviews the software detailed design documents, assigns configuration items according to the defined plan, and updates the configuration item

- index. SCM also releases the software detailed design documents and places them under change control.
5. *Product*—SCM reviews the developed software to ensure that all programs are fully updated and have been tested and released to test and evaluation.
 6. *Operational*—SCM ensures that all software components are fully updated and have passed appropriate acceptance tests.

7.4 Standards for Configuration Audit Functions

We'll discuss the standards here in three parts; ISO, CMM and IEEE.

In the older ISO standard (ISO 9000:1994) & ISO 9000-3 :1997 (Guidelines for application of ISO 9001 to the development, supply and maintenance of software), the clauses that directly address Configuration Management are:

- Clause 4.5 Document and Data Control
- Clause 4.8 Product Identification and Traceability
- Clause 4.12 Inspection and Test Status
- Clause 4.4.9 Design Changes (Design Control)
- Clause 4.13 Control of Non-conforming Products

Corresponding clauses (to those above from the older ISO standard) exist in the new ISO 9001:2000 standard as shown in the table below: We'll not provide here the full text of each clause. Interested readers are directed to reference [5]

Table 3: SCM and the ISO Standard (Old and New)

Clause from Old ISO Std (ISO 9000:1994)	Title	Clause from New ISO Std (ISO 9001:2000)	Title
4.5	Document and Data Control	4.2.3	Control of Documents
4.8	Product Identification and Traceability	7.5.3	Identification and Traceability
4.12	Inspection and Test Status	7.5.3	Identification and Traceability
4.4.9	Design Changes	7.3.7	Control of Design and Development Changes
4.13.1, 4.13.2	Control of Non-conforming Products	8.3	Control of Non-conforming Products

The details of the CMM are covered in a separate chapter. As far as CMM (Capability Maturity Model) is concerned, the following can be said in brief:

- Purpose
 - To establish and maintain the integrity of the products of the software project throughout the project's software life cycle
- Involves
 - Identifying configuration of software
 - Systematically controlling changes to configuration
 - Maintain integrity and traceability of configuration management
- Work Products under configuration Management
 - Software products to deliver
 - Items identified with or required to create them

CMM provides some goals pertaining to SCM; these are stated below:

- GOAL 1—Software Configuration management activities are planned
- GOAL 2—Selected software work products are identified, controlled, and Available
- GOAL 3—Changes to identified software work products are controlled
- GOAL 4—Affected groups and individuals are informed of the status and content of software baselines

CMM looks for the following *Commitments, Abilities and Activities* towards SCM in an organization:

Commitment

- COMMITMENT 1—The project follows a written organization policy for implementing software configuration management (SCM)
 - Responsibility explicitly defined
 - SCM throughout life cycle
 - For external and internal work products and designated support tools
 - Repository
 - Baselines audited periodically

Ability

- ABILITY 1—A board having the authority for managing the project's software Baselines (i.e. a software configuration control board exists or is Established)
- ABILITY 2—A group that is responsible for coordinating and implementing SCM for the project (i.e. the SCM group) exists
- ABILITY 3—Adequate resources and funding are provided for performing the SCM activities

- **ABILITY 4**—Members of the SCM group are trained in the objectives, procedures and methods for performing their SCM activities
- **ABILITY 5**—Members of the software engineering group and other software related groups are trained to perform their SCM activities

Activities

- **ACTIVITY 1**— A SCM plan is prepared for each software project according to a documented procedure
- **ACTIVITY 2**— A documented and approved SCM plan is used as the basis for performing the SCM activities
- **ACTIVITY 3**— A configuration management library system is established as a repository for the software baselines (may have multiple control levels of SCM)
- **ACTIVITY 4**— The software work products to be placed under configuration management are identified
- **ACTIVITY 5**— Change requests and problem reports for all configuration items/units are initiated, recorded, reviewed, approved, and tracked according to a documented procedure
- **ACTIVITY 6**— Changes to baselines are controlled according to a documented Procedure
- **ACTIVITY 7**— Products from the software baseline library are created and their release is controlled according to a documented procedure
- **ACTIVITY 8**— The status of configuration item/units is recorded according to a documented procedure
- **ACTIVITY 9**— Standard reports documenting the SCM activities and the contents of the software baseline are developed and made available to affected groups and individuals
- **ACTIVITY 10**— Software baseline audits are conducted according to a documented procedure. It should be noted that here it is typically done in organizations by using standard checklists developed for configuration audit.

The CMM model also provides Measurement and Verification towards Configuration Management activities; however they are not discussed here. Interested readers can study the CMM model by

Visiting the site www.sei.cmu.edu

Now let us discuss about the standards produced by the Institute of Electrical and Electronic Engineers (IEEE). In 1976, a subgroup was formed under IEEE to develop a software quality assurance standard.

This has resulted in over 20+ standards and guides for related to software. The ones related to Software Configuration Management are:

- IEEE-SCM 828-1990 : Standard for SCM Plans
- IEEE-SCM 1042-1987 (r: 1993): Guide to SCM

IEEE standards do not take you towards 'certification' but this standard can be used to implement the requirements of ISO or CMM for SCM. Here is a brief overview on both the IEEE standards mentioned above:

IEEE-SCM 828-1990: Standard for SCM Plans

- Introduction
 - Purpose of the plan, scope of the SCM plan, intended use of the SCM plan, overview of the project, SCM issues and risks in project context, degree of SCM formality, limitations, assumptions, terminologies, references to other documents
- SCM Management
 - Software Project Organization structure, roles and responsibilities, directives and policies, references to (organization's or customer's) procedures and standards to be followed (in the context of SCM)
- SCM Activities
 - Configuration items identification, change management and configuration control, status accounting, configuration audits, interface and subcontractor control
- SCM Procedures
 - Sequence of SCM events, dependence between SCM activities, relationship to other key project milestones
- SCM Resources
 - Personnel, Tools, techniques, Equipments, Training, use of other personnel
- SCM Plan Maintenance
 - Responsibility, conditions for update, approvals communication of changes

IEEE-SCM 1042-1987 (r:1993):Guide to SCM

- Provides guidance in planning software configuration management practices that are compatible with IEEE 828. In Table 4 below we have provided a checklist. It can be used for evaluating the scope of system under consideration for the preparing Software Configuration Management Plan (SCMP)

Table 4: Checklist for Evaluating the Scope of the System for Software Configuration Management Plan

- Are the boundaries the system to be audited clearly delineated?
- Have these definitions been made a part of the systems specification?
- Has the purpose of the system and the function of the computers been clearly defined?
- Have the criteria for systems requirements allocation to the various configuration items been clearly established and made public?
- Is the Configuration Management system to be audited clearly defined?
- Have all tools to be used for the management of the system's configuration been put into place, or has a clear schedule been allocated for such?
- Does an organization exist for the performance of the configuration Management duties?
- Are the Configuration Management audits defined publicly?

- It is intended for software developers, software management community, those responsible for preparing SCM plans
- The guide is structured as two parts:
 - o Main body presents overview, principles, issues, lessons essential concepts, etc.
 - o Second part contains 4 sample plans
 - For critical software for embedded systems
 - For experimental development of small systems
 - For a software maintenance organization
 - For a product line system

Figure 3 presents a snapshot for IEEE standard towards Software Configuration Management.

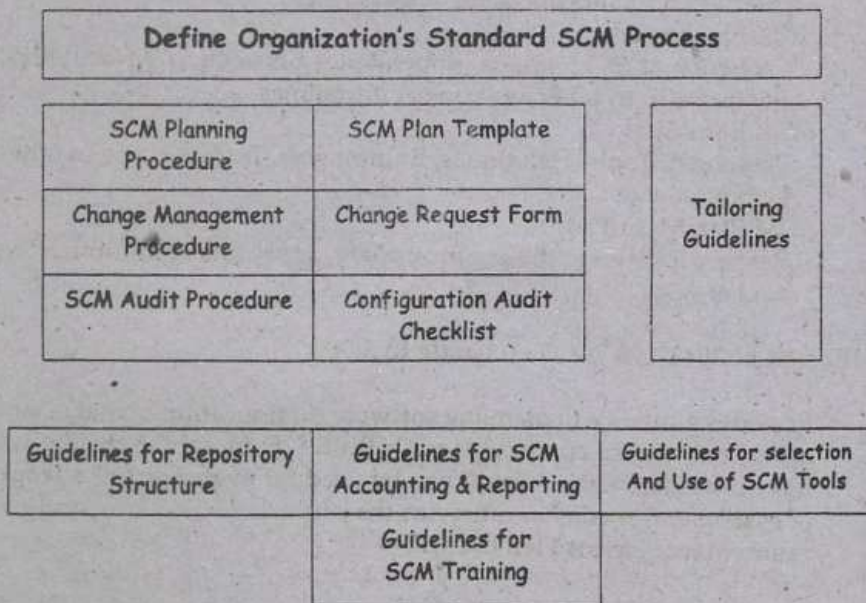


Figure 3: IEEE: SCM Snapshot

7.5 Personnel in SCM Activities

As in any management discipline, it is critical to question *who is to perform the task of configuration management*. Characteristics of personnel needed for efficient performance of Software The organization must give close attention to Configuration Management tasks. This section describes some of this information in a summary manner. In principle, there are five information items concerning the 'people-ware' that must be examined in detail.

1. The number of personnel needed for each software configuration management task (the tasks of SCM have been discussed in the previous sections of this chapter)—this will form an important consideration for project budgeting. Usual rule of thumb is that, once the system has been set up, we should need one person per project group, for about five percent of their professional time on core project job.
2. The expected skill levels for all those involved—the professional to set up the system up must be at a very high skill level. The person to run the configuration management system can be at a much lower level.
3. The job titles of those who perform the tasks are significant but these can vary depending on the project organization, degree of formality required, which in turn is dictated by the scope and complexity of the software project at hand.
4. Geographic locations—are all the people co-located at one geography/city/building/site, or are some of them located at a remote site. While using a client/server based SCM tools this factor plays a critical role in setting up the computer-based Configuration management tools.
5. Are any special security clearances needed? This does not only refer to 'military' kinds of security! Many commercial organizations have security issues to deal with (they have their internal audit programs to audit those who audit!). Certainly, all organizations must, at very least, have back-ups to prevent loss in case of hardware failure.

7.6 Software Configuration Management: Some Pitfalls

So far, we have discussed the ideal situations for Software Configuration Management i.e. how things should be with respect to SCM. However, life is not known to be simple and as a result, SCM situations to transgress from 'ideal' to something far less than ideal. Of course, like for most things in life, many factors contribute to this and the idea in this section is to discuss them so that we can understand how to avoid them. These anti-patterns can be summarized as

indicated below. A full discussion on each of them is not within the scope of this book; the interested readers can visit reference [6]. Before that, let us have a quick discussion on the notion of 'anti-pattern'. Anti-Pattern is defined as "a commonly occurring solution to a problem that generates decidedly negative consequences; in other words, an Anti-Pattern is generally a commonly repeated practice in industry that is flawed in some manner and that causes symptoms that are more undesirable than whatever positive results were sought. In sum, 'anti-patterns' are the syndromes to be watched for.

- *Silver Bullet*—the mistaken belief that a tool can solve all your configuration management problems.
- *CM Takeover*—situation where configuration management becomes an all-consuming task; an end unto itself.
- *Developer-Software Configuration Management*—what happens when configuration management is left to the development team rather than an independence configuration management organization.
- *Decentralized Configuration Management*—insight into the value of centralized repository for performing development and configuration management.
- *Failure to Audit*—team fails to implement one of the fundamental aspect of configuration management; performing configuration audit.
- *Object Oriented Configuration Management*—Criticality of configuration management to object-oriented projects, which have varying levels of granularity and degree of complexity introduced by object-oriented development.
- *Software Configuration Management Expert*—those instances where unqualified individuals are given project configuration management responsibilities, and who don't know that they are not qualified.
- *Postmortem Planning*—importance of planning, and follow-through for configuration management, and results you get if you don't.

Conclusion

Software Configuration Management (SCM) is an all too important activity throughout a software project. When the project is large and complex, it requires an SCM control over requirements and specifications to ensure that the product is build and tested as per expressed requirements of the customer. SCM is an umbrella activity and as such, it should be maintained over the design of an application throughout the software development life cycle (SDLC) to ensure its integrity and maintainability. For this, requirements baselines, specification baselines, design baselines etc are established and maintained at appropriate points throughout the development cycle.

As a first step towards establishing the SCM, a software configuration management plan should be developed. Project naming conventions are established and a family of forms and procedures is provided to ensure that every change is recorded, reviewed and tracked. The software requirements specification (SRS) document is used as a basis for the design and development work and further up to acceptance testing stage.

During the design phase, SCM maintains control over the design. As changes are made, the appropriate design document is updated correspondingly. SCM facilities are also required to record all change to the code. For large projects and for all projects during the maintenance phase, procedures are needed to handle the development of simultaneous versions of the same program. The tools used to design, implement, test and maintain the software must also be maintained under configuration control as these, two may undergo versions change; especially in case of long running projects.

Software Configuration status accounting is aimed at maintaining a continuous record of the status of all base-lined items. Software configuration audits are periodically performed to ensure that the SCM practices and procedures are followed as per SCM plan.

REFERENCES

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- [3] *The Capability Maturity Model: Guidelines for Improving the Software Process* (Carnegie Mellon University—Software Engineering Institute)
- [4] *CMM in Practice*—Pankaj Jalote (Addison Wesley)
- [5] *Implementing ISO 9001:2000: The Journey from Conformance to Performance*—Tom Taormina (Prentice Hall PTR)
- [6] *Anti-Patterns and Patterns in software Configuration Management*—William Brown (Wiley Computer Publishing)

ISO 9001

8.1 Overview

In Chapter 2, we had an extensive discussion on Quality (Management) System—the QMS. ISO 9001 is a well-conceived *standard* to achieve continual improvement. In normal language, *quality* is often linked to an idea of “excellence”. Excellence can be achieved through continuous focus on “process improvement. In Chapter 9, we have discussed, a number of models for Process Improvement. ISO 9001:2000 is mentioned there. In this chapter, our focus is only ISO 9001.

The discussion in this chapter is broadly divided into three parts; the structure of ISO, its working (these are the lesser known details of ISO) and its brief history. Then there is a discussion about the 2000 revision of ISO giving an overview of the clauses in this new revised ISO 9001:2000 standard. Preparing for ISO Audits/Assessments and use of consultants etc. is also discussed briefly.

A couple of points to be noted—although many people believe this to be so, the word “ISO” is not the acronym for the organization but is derived from the Greek word “isos” meaning equal; as in isobars (equality of laws), isonomy (people before the law). Often ISO 9000 and TQM get mentioned together. The concept of delighting customers leads into *Total Quality Management (TQM)*, which is considered a step beyond what can be achieved by putting in an effective *quality management system (QMS)* such as through use of ISO 9000 as a framework for *process improvement*. However, the difference between fully meeting customer requirements and delighting them, is in practice slight, possibly more semantic than real.

8.2 What is ISO 9000

The International Organization for Standardization (ISO) is a worldwide federation of national standards bodies, one from each from over one hundred and forty countries. The table below shows the details regarding member constitution of ISO. (The data is as of January 2003).

Table 1: Member Constitution of ISO

Members	146	National standards bodies, comprising
	94	Member bodies (P-Members)
	37	Correspondent members (O-Members)
	15	Subscriber members
Technical Committee Structure	2 937	Technical bodies, comprising
	188	Technical committees
	550	Subcommittees
	2 175	Working groups and
	24	Ad hoc study groups
Staff Technical	36	Member bodies provide the administrative and technical services for the secretariats of technical committees (TC) and sub-committees (SC)
	500	These services equal a full-time staff of Persons
Secretariats Central Secretariat in Geneva	163	Full-time staff from
	25	Countries coordinate the worldwide activities of ISO
Financing	140	Million CHF per year is estimated as the operational expenditure for the ISO work, of which
	80%	Is financed directly by
	36	Member bodies holding TC and SC secretariats, and
	20%	Through member body subscriptions and publications income, covering the costs of the Central Secretariat

ISO is a non-governmental organization, which sees its mission as the promotion of the development of standardization and related activities in the world. Its work results in international agreements, which are published as International Standards. Figure 1 shows the structure of ISO.

ISO is made up of three categories of members as you can see in Table 1 above. *Member Body of ISO* is the national body most representative of Standardization in its country and as such there is only one such body admitted to membership. They have full voting rights and are known as "P-Members".

Correspondent member is usually an organization in a country, which does not yet have a fully developed national standards activity. They do not participate in policy or development work but are entitled to be kept fully informed concerning relevant issues. They are known as "O-Members".

Subscriber membership is for countries with very small economies and as a result they pay reduced membership fees. However, this membership does enable them to maintain contact with international standardization work.

8.3 The Origins of ISO 9000

The origins of ISO 9000 lie in wartime and the defense industry. The problem of defects in ammunition and the need for reliability in equipment on which survival might depend, moved the focus to

controlling how the work was done. Eventually this led to defined standards for quality management to which defense contractors were expected to work. As the concept of *quality management* and *quality systems* became known, buyers in non-defense industries started to demand that their own suppliers of technical products adopt quality systems.

Thus, it all started in the electro technical field and its forerunner was the IEC—the International Electro-Technical Commission, which was created in 1906. In addition, the International Federation of the National Standardizing Associations (ISA), which was created in 1926, undertook other pioneering work, primarily in mechanical engineering.

The Second World War interrupted their work and subsequent to a meeting of twenty-five countries in 1946, a new international organization was created with the objective of "the facilitation of the international co-ordination and unification of industrial standards". With this, ISO was duly formed and began its official functions on 23rd February 1947.

8.4 How Does ISO (as an organization) Carry Out Its Work?

The work of ISO is carried out by a hierarchy of "technical committees (TCs) sub-committees (SCs) and working groups (WGs) as shown in Figure 1.

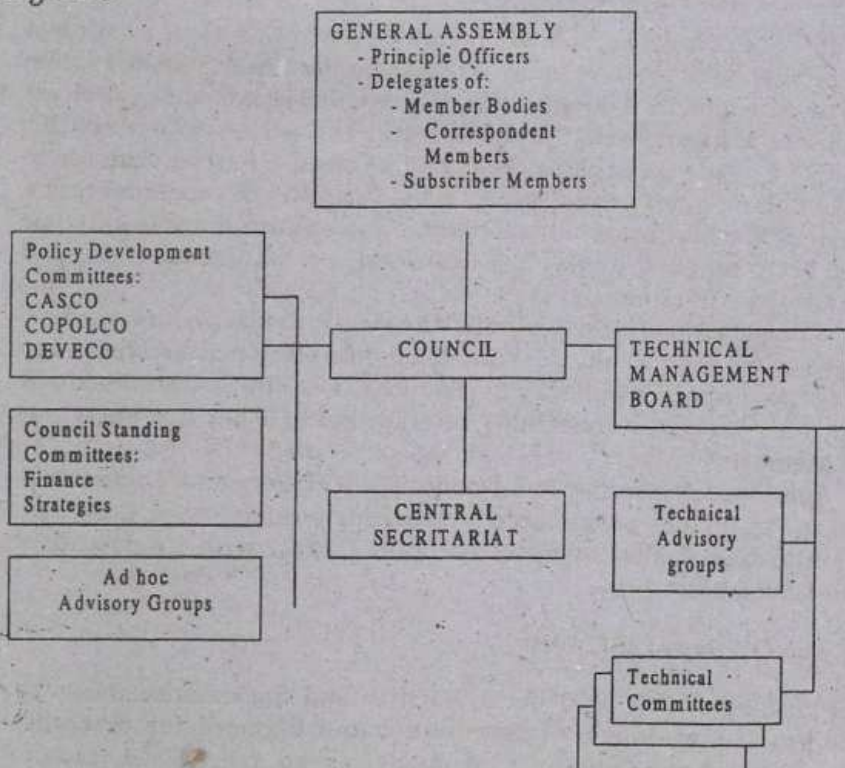


Figure 1: ISO Structure

All these TCs, SCs and WGs total up to more than 2500 groups. With these committees there are qualified industry representatives, research institutes, consumer bodies, government authorities and world wide international organizations whose objective is the resolution of global standardization issues. They are all of equal status within a group and it is estimated that more than 30,000 experts participate in meetings each year.

The Central Secretariat of ISO, located in Geneva, co-ordinates the dissemination of documentation; facilitates clarification with chairmen and the agreement of technical content by committees; co-ordinates meetings and generally provides overall support.

ISO issues a publication, *ISO Memento*, which provides information on the scope of responsibility, organizational structure and secretariats for each ISO Technical Committee. Detailed rules of procedure for technical work are given in ISO/IECC directives and are beyond the scope of discussion in this chapter.

8.5 ISO Standards Development Process

An International Standard is the result of an agreement between the member bodies of ISO through the development process of the technical committees (TCs) and the sub-committees (SCs). The process has six stages as shown below:

- Stage 1 Proposal Stage
- Stage 2 Preparatory Stage
- Stage 3 Committee Stage
- Stage 4 Enquiry Stage
- Stage 5 Approval Stage
- Stage 6 Publication Stage

Brief details on each of the stages mentioned above are provided below.

Stage 1 (Proposal)

The first step is to ascertain the need for a proposed standard. To evaluate the proposed standard, a new work item is submitted for vote by the members of the relevant TC/SC to determine if the item is to be added to the work program. The proposal is accepted only if a majority of the P-Members (refer section 8.2) of the relevant TC/SC vote in favour and that at least five P-members declare that they are committed to participate in the project actively. This normally results in the appointment of a project leader.

Stage 2 (Preparatory)

Typically, a group of experts, together with the project leader (appointed at the end of Stage 1), is established and they prepare a working draft. Successive drafts are potentially possible until the group is satisfied that the best technical solution has been developed. At this point, the draft is released as a "Committee Draft" and passed to the group's parent committee for the next consensus building stage.

Stage 3 (Committee)

Upon creation of the first draft it is registered by the ISO Central Secretariat (refer Figure 1) and distributed for comment by P-Members of the TC/SC. Successive committee drafts may also be considered until consensus is reached on the technical content. Once the consensus has been reached, the text is finalized for submission as a "Draft International Standard" (DIS).

Stage 4 (Enquiry)

The DIS (Draft International Standard) is circulated to all the ISO member bodies by the ISO Central Secretariat for voting and comment within a period of five months. It is approved for submission as a "Final Draft International Standard" (FDIS) if a two-thirds majority of the P-members of the TC/SC are in favour and not more than one quarter of the total number of votes cast are against. In the event that the text is unapproved, it is returned to the originating TC/SC for further evaluation and study and revised document is then subjected to the above approval process.

Stage 5 (Approval)

The FDIS (Final Draft International Standard) is circulated to all ISO member bodies by the ISO Central Secretariat for a final Yes/No vote within a two month period. If technical comments are received at this stage, they are registered and held over to the next revision of the International Standard. If two thirds of the P-members of the TC/SC are in favour and not more than one quarter of the total number of votes cast are against, the FDIS text is approved. However, if these criteria are not fulfilled, the standard is referred back to the originating TC/SC for re-consideration of the technical reasons submitted supporting the negative votes.

Stage 6 (Publication)

Once the FDIS has been approved, no other amendments are allowed other than minor editorial changes. The text is then passed to the ISO Central Secretariat who then publish it as the "International Standard".

3.6 How the ISO 9000 Family of Standards Work

With this background, let us now understand how the ISO 9000 family of standards works. Although the requirements for a quality system are standardized—most of us like to think our business is unique. So how does ISO 9000 allow for the diversity of say, on the one hand, a small enterprise, and on the other, to a multinational manufacturing company with service components, or a public utility, or a government administration? The answer is that ISO 9000 lays down *what* requirements your quality system must meet, but does not dictate *how* they should be met in your organization—which leaves great scope and flexibility for implementation in different business sectors and business cultures...as well as different national cultures. Thus, ISO 9000 does not take a 'prescriptive' approach; it is a 'normative' standard.

The ISO 9000 family includes standards that give organizations guidance and requirements on what constitutes an effective quality management system. ISO 9004-1 (and the other parts of ISO 9004) are the standards giving guidelines on the elements of quality management and a quality system. The family also includes models against which this system can be audited to give the organization and to its clients assurance that the system is operating effectively. As with the previous version (ISO 9000:1994), three quality assurance models are ISO 9001, ISO 9002 and ISO 9003 (the new revision is discussed in the next section). Lastly, the family includes a standard on terminology, and other standards, which can be described as "supporting tools", that give guidance on specific aspects, such as auditing quality systems.

3.7 ISO 9001:2000

Figure 3 in Chapter 2 shows the diagrammatic overview of the concept behind the 2000 revision of ISO. Readers may find it of interest to know that the first revision to ISO 9000:1987 came in 1994. This resulted in ISO 9000:1994. It was a combination of minor enhancements and classification of ambiguities. The year 2000 revision (known as the ISO 9001:2000) is a "total re-write" of the 1994 standard. The twenty elements of the ISO 9000:1994 (some people refer to them as the "20 clauses") have been replaced by the clauses summarized below.

- Quality Management System (with 2 sub-clauses within it)
- Management Responsibility (with 6 sub-clauses within it)
- Resource Management (with 4 sub-clauses within it)
- Product Realization (with 6 sub-clauses within it)
- Measurement, Analysis and Improvement (with 5 sub-clauses within it)

(For a detailed mapping between the older ISO 9000:1994 and the new ISO 9001:2000 standard readers can refer to ref [3].

It should be noted that although all 20 of the elements/clauses of the older version of ISO (ISO 9000:1994) are contained in the 5 clauses of the current version of the ISO Standard (ISO 9001:2000), there are substantial changes and new requirements have been laid down. They include the following (Major and Minor changes):

- *Elimination of ISO 9002 and ISO 9003*
- The ability to exclude sections of ISO 9001 that do not apply to your organization
- Redefinition of ISO 9000 to "Fundamentals and Vocabulary", which eliminated ISO 8402
- Redefinition of ISO 9004 to "Guidelines for Performance Improvements"
- Commonality of the formats of ISO 9000, ISO 9001, and ISO 9004
- Change from "Quality Assurance System" to "Quality Management System" (QMS)
- Encouragement of using "the *process approach*" (Ref Figure 3 in Chapter 2)
- The use of the term "organization" rather than "supplier"
- The use of term "records" instead of "quality records"
- *Enhancing the scope of the term "Product"* to mean hardware, software, services, and processed materials.
- Reduction in the number of documented procedures required (from 18 to 6)
- Slight increase in the number of required records (from 17 to 21)
- Identification of processes, their sequences, and interaction
- Establishing the need for criteria and methods for effective operation and control of processes
- Evidence of management commitment to QMS
- Establishment of *measurable quality objectives*
- Opening channels of *internal communications*
- Addressing and measuring *customer satisfaction* with focus on meeting customers' needs and expectations
- Identifying *competency needs*
- Evaluating *training effectiveness*
- Making employees aware of how their work affects the QMS
- Managing the *human and physical work environment*
- Communicating awareness of customer requirements
- Communicating with customer at all levels
- Expanding monitoring and measuring activities to run the QMS
- Controlling processes as well as products
- Expanding statistical techniques to *data analysis* and taking action on the data

- Building a *continual improvement* model

Thus, in summary, the ISO 9001:2000 is now structured in following sections (Refer Figure 2.)

- Foreword
- 0 Introduction
- 1. Scope
- 2. Normative Reference
- 3. Terms and definitions
- 4. Quality Management System
- 5. Management Responsibility
- 6. Resource Management
- 7. Product Realization
- 8. Measurement, analysis and improvement

Those who are quite familiar with the older version (ISO 9000:1994) may like to note that sections 4-8 (indicated above) contain the clauses 4.1 to 4.20 together with additional requirements (indicated through the major and minor changes already discussed). With this in the background, let us discuss the organizational need for the ISO Standard.

3.8 Why do Organizations Need ISO 9000?

The ISO 9000 family of standards represents an international consensus on good management practices with the aim of ensuring that the organization can time and time again deliver the product or services that meet the client's quality requirements. These good practices have been distilled into a set of standardized requirements for a quality management system (QMS), regardless of what your organization does, its size, or whether it's in the private, or public sector.

There is a business case for the need for the ISO Standard. Often, people ask 'Why Should my organization implement ISO ? The answer is really simple! The existence of an organization without customers, or with dissatisfied customers, is in peril! To retain customers—and to keep them satisfied—your product (which may, in fact, be a service) needs to meet their requirements. ISO 9000 provides a tried and tested framework for taking a systematic approach to managing your business processes (your organization's activities) so that they consistently turn out product conforming to the customer's expectations. And that means consistently happy customers! This aspect of ISO is represented well in Figure 3 of Chapter 2. (We have repeated this figure below for readers' quick reference—see Figure 2 on the next page)

8.9 ISO Certification

8.9.1 Importance of the Certification

The organization should carry out auditing of its ISO 9000-based quality system itself to verify that it is managing its processes effectively—or, to put it another way, to check that it is fully in control of its activities. In addition, the organization may invite its clients to audit the quality system in order to give them confidence that the organization is capable of delivering products or services that will meet their requirements.

Lastly, the organization may engage the services of an independent quality system certification body (the Registrars) to obtain an ISO 9000 certificate of conformity. This last option has proved extremely popular in the market place because of the perceived credibility of an independent assessment (the topic of assessment has been covered in great details in a later section of this chapter). It may thus avoid multiple audits by the organization's clients, or reduce the frequency or duration of client audits.

The certificate can also serve as a business reference between the organization and potential clients, especially when supplier and client are new to each other, or far removed geographically, as in an export context. In a way, it is a risk-mitigation exercise (the risk is not being able to meet quality requirements in the product/service to be purchased by the customer and to be supplied by the vendor.)

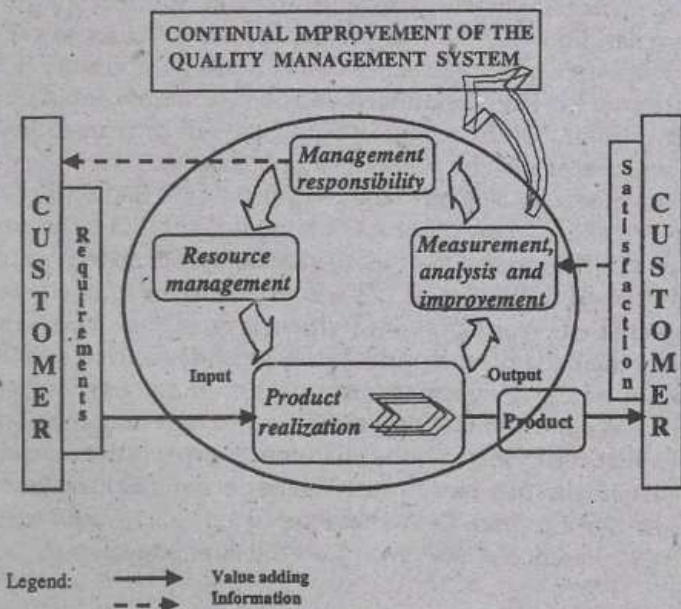


Figure 2: Conceptual Presentation of the QMS Model

8.9.2 Getting Ready for ISO Certification

From the discussion so far in this chapter, and Chapter 2, it should become apparent that the quality (management) system or the QMS requirements and *procedures* (they are the heart of the QMS) developed to implement them all closely lock together. The quality triangle illustrated in Figure 3 represents this concept.

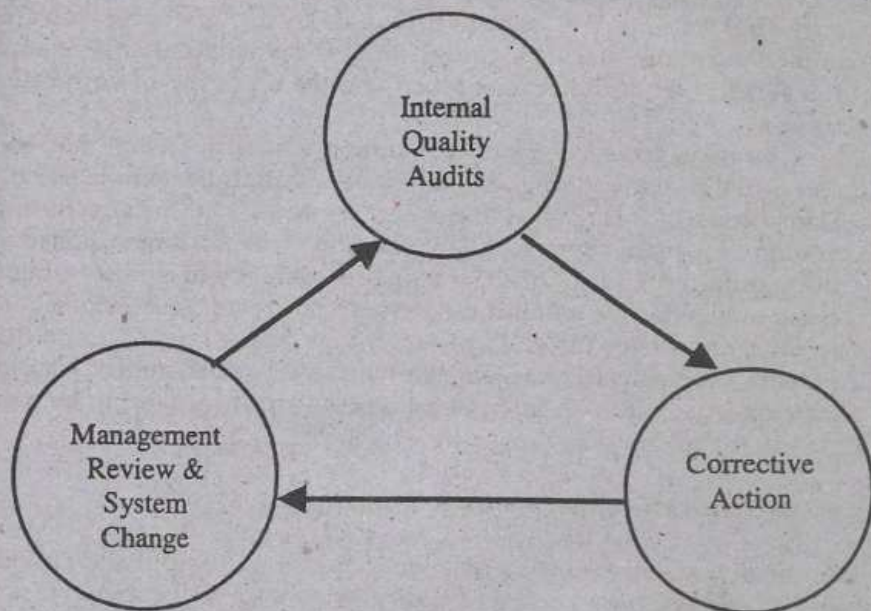


Figure 3: The Quality Triangle

Clause 8.2.2 of ISO 9001:2000 is stated below. As per that, a process for internal (quality) audit is required. Such audits must be carried out by trained staff (suitable training can be provided internally).

Clause 8.2.2: Internal Audit

The organization shall conduct internal audits at planned intervals to determine whether the quality management system

- conforms to the planned arrangement to the requirements of this International Standard and to the quality management system requirements established by the organization and*
- is effectively implemented and maintained.*

An audit program shall be planned, taking into consideration the status and importance of the processes and areas to be audited, as well as the results of previous audits. The audit criteria, scope, frequency and methods

shall be defined. Selection of auditors and conduct of audit shall ensure objectivity and impartiality of the audit process. Auditors shall not audit their own work.

The responsibility and requirements for planning and conducting audits, and for reporting results and maintaining records shall be defined in a documented procedure.

The management responsible for the area being audited shall ensure that actions are taken without delay to eliminate detected non-conformities and their causes. Follow-up activities shall include the verification of the actions taken and the reporting of verification results.

The management personnel planning a quality system may well feel initially that auditing appears to be a dull, bureaucratic activity. Done properly, it is none of these, and is essential for the effective operation of a quality system. Even if auditing was not a requirement of the Standard (ISO 9001:2000), any quality system will almost certainly cease to be effective without this important process. Furthermore, internal audits mirror the work of external assessors/auditors and thus prepare a company for the final (external) assessments/audits. The only difference is that often-internal auditor's work may tend to be more thorough and 'tougher'.

8.9.3 Certification, Assessment and Audit

In the discussion above we have used the terms "audit" and "assessment". We also have made a reference to the term *certification*. We now spend some time of these terms. Often people ask if there is any difference between the terms *Assessment* and *Audit*. Various opinions exist on this. In one of early works on software assessment, Roger Pressman (the author of the well known book "Software Engineering: A Practitioner's Approach") defines the *assessment* as 'a structured investigation of the current state of the software engineering practice'. Pressman relates the two terms by stating that *audit* is a primary mechanism for the *assessment*.

The IEEE has a specific definition of an audit. 'Audit—An independent examination of a work product or set of work products to assess compliance with specifications, standards, contractual agreements or other criteria' (IEEE STD-610). In reality, the term 'audit' is associated with more formal inspections and certifications performed by external resources (like a financial audit). Using the word 'audit' to refer to *software process assessment* can create a mental barrier between the assessment team and the assessment participants.

ISO documents define the concepts of *certification* and *audit* as follows: *Certification or third-party assessment* (referred to as 'registration' in some countries—including India), is carried out by an independent

organization against a particular standard (ISO being one of them). Watts Humphrey's view (he is the author of the world famous book "Managing the Software Process") is that 'a software process assessment is not an audit but a review of a software organization to advise its management and professionals on how they can improve their operation.

There are some more expert views on these terms [4]. 'Certification' is the procedure by which a third party gives written assurance that a product, process or services conforms to specified characteristics. 'Assessment' is the process of comparing the actual *measurements* (topic of measurements is discussed in detail in Chapter 5) of the characteristics of interest with the specifications of those characteristics. The most common perception of 'certification' is that it gives some form of 'guarantee' generally of quality and dependability in their widest senses. At least it is assumed that the risks of failure are reduced because certification involves some form of independent assessment.

Having discussed the terms—audit assessment and certification; let us now turn to the various types of audits and assessments.

8.9.4 Types of Assessments

Generally, there are three different types of assessments depending on who plays the main role in an assessment.

- *Self-Assessment*: Also known as 'first-party assessment'. This refers primarily to a situation where the assessment is performed internally inside the software development organization; mainly by its own personnel. The main objective is to identify the organization's own *software process capability* and initiate an action plan for *software process improvement*. First party assessment can also be performed as guided first-party assessment where external assessors act as assessment methodology facilitators and coaches for the internal assessment team.
- *Second-party Assessment*: Also known as 'capability determination'. This is an assessment where external assessors are used to perform the assessment. The main objective is to evaluate the organization's capability to fulfill specific contract requirements.
- *Third-party Assessment*: Also known as 'capability determination'. This is performed by an independent third-party organization. The main objective is to verify the organization's ability to enter contracts or produce software products, and sometimes to provide the fulfillment of certification according to a selected standard.

8.9.5 Types of Audits

The classification provided below is consistent with the British Standard Institution's classification of an audit.

- **First-party Audit:** When an organization carries out an *internal audit*, the procedure is referred to as '*first-party audit*'. This is done by organizations to check its quality procedures before any third party is invited to carry out a formal *compliance audit* (compliance means adhering to something specified as standard). An internal audit is equivalent to self-assessment.
- **Second-party Audit:** A *second-party audit* is an audit of an organization's relevant products and processes by a customer (acquirer). The product or quality system standards are typically selected by the customer and specified as a part of the contract. Second-party audits may be relatively narrow in scope because the purchasing organization examines only those elements and activities that relate to the products and services being purchased/acquired. The pre-contract survey is an example of a second-party audit.
- **Third-party Audit:** This is an audit of an organization at its own request and expense carried out by an external, impartial body referred to as a *certification body*, which is not a direct customer. The third-party audit is invariably performed against the requirements of a national or international standard such as ISO 9001.

8.10. Assessment/Audit Preparation

A quality (management) system cannot be realistically assessed until it has operated for some time. The assessment process involves the 'hard' evidence of completed records required by the procedures/processes. It should be noted that assessment, like 'internal auditing' can also include observing and talking to the staff. To build up an adequate body of such data takes time and until this is sufficient, a meaningful assessment cannot be carried out. It is for this reason that most assessment bodies will insist on a certain preparatory period. Sometimes the consultants play this role of assessment body. (The next section has discussion on the role of ISO consultants)

8.10.1 Gestation Period

How long is this period? Most assessors/consultants are likely to require a minimum of about three months, but longer may well be appropriate, depending on many aspects of the aspirant organization, its internal structures, business processes, nature of products/services etc. There are arguments against being 'over-cautious' and waiting too long for assessment. (Readers should keep in mind the "types of assessments" discussed in section 8.9.4)

8.10.2 Importance of Internal Audit

Internal auditing is especially vital from this perspective. The aim of internal audit should be to cover the whole system between implementation

and final assessment. In this way, problems and gaps should be identified and dealt with before an assessor picks them up. In short, the importance of effective internal auditing cannot be over-emphasized.

An option before full assessment is a pre-assessment audit. This is in effect a dummy run of the real thing and unlike routine internal audits, it is carried out by outsiders. An ISO consultant involved (much more on this in the next section) throughout the project (of getting the organization ISO certified) can be considered for this work. However, it should be noted that if the same individual is involved, certain "independence" is lacking. Although there are many freelances and other independent organizations offering this service, (as we'll see in section 8.13.1) the obvious choice for pre-assessment is the external body that will carry out the final assessment. This way, at least you learn how the particular body carries out the assessment process.

Pre-assessment is particularly appropriate for organizations with little or no outside help. In such cases, it (the pre-assessment) may identify areas of the quality (management) system which are weak, or do not fully match up to ISO 9001 requirements.

8.10.3 Cost and other strategic considerations

If an organization is concerned to keep costs down then it as well may trust its own judgment on the effectiveness of its QMS (quality management system) and go straight for the final assessment. If the first attempt is unsuccessful, then little is lost since the money saved (assuming that the organization did not hire external consultant help!) will cover a second full assessment. However, there is a counter to this argument too—what about the staff de-motivation upon having "failed" the assessment and also the top management disappointment as well as "loss of faith"? Also, a fact remains that a pre-assessment may have a larger "diagnostic" element than a full assessment (where how to put right a problem may not get fully addressed). Finally, another argument against spending money on pre-assessment is that it may be better to invest more in training your own staff as internal auditors who will then be able to carry out their own equivalent of a pre-assessment. With this approach, the benefits are long term; your own trained people are going to be with you (assuming no attrition!)

8.11 The Assessment Process

Once a contract is signed with the chosen body, the details of the assessment process will get agreed with the management representative—the MR (person appointed by the client organization to liaise with the external assessors). Now the schedule for the assessment gets decided. On-site assessment is the second stage of this process

and in the intervening period, a "desk investigation" is carried out.

Assessors may vary in their approach to and detailing of work carried out during desk investigation. Some do it off-site and others make a "special visit". Whatever the approach, the objective is always restricted to establishing whether the documented quality system meets the requirements of the ISO Standard (indicated in the 8 clauses—refer end of section 8.7). As a result of the desk investigation, the assessor may very well raise a number of queries and these must be dealt with. The MR (Management Representative) is involved all through and the discussions with the assessor should resolve what is required. If the assessors consider that the QMS (quality management system) is not up to the mark (as demonstrated through the major gaps found during desk investigation), then there is no point in going for the final assessment until these are put right.

The on-site assessment process is in principle the same as internal audit work (assuming it is done properly) except that the whole QMS is covered at one go. The duration of on-site assessment would depend on the manpower allocated by the assessors and the size and complexity of the client organization's business operations, software life cycles etc. During the on-site assessment, a systematic checking is done whether each element of the QMS is being implemented effectively and the procedures/processes followed. This is done through the method of "internal audit" mentioned earlier (section 8.9.5). The areas typically examined are records of internal auditing, corrective actions, and management review, including a check on changes to the QMS since the desk investigation.

Most organizations and their staff are likely to find assessment an ordeal however well prepared they are. Although generally the assessment body's outlook is positive (they want to find the evidence to award ISO certification), non-conformities (NCs) cannot be ignored. Non-conformities raised are of two kinds—*Minor NC* and *Major NC*. Minor Non-conformities are some defects in the Quality Management System (for example, some records are found to be incomplete while the majority are proper). Major non-conformities are a serious matter. These are raised if the assessor considers that a whole area of the QMS (quality management system) is not being implemented effectively. Also, a number of Minor Non-conformities may be considered to add up to a major one, particularly if a consistent theme in them is apparent. The consequence of Major Non-conformity is that ISO certification cannot be awarded until a further re-assessment visit (at an extra cost) has been made. Under such situations, organizations do feel a great disappointment but it would be wrong to dwell on this as a permanent failure—remember the whole journey is supposed to be towards "process improvement".

8.12 Surveillance Audits/Re-Certification/Re-Assessment Audits

If all goes well at the assessment visit, the organization will be awarded ISO certification. Mostly the assessors/external-auditing agency (who visited you for the final assessment) would announce this in the closing meeting. However, there is often a few weeks' delay until the certificate arrives and it is official. This is when the 'joyous' moment arrives and organizations go in spate of public announcements and celebration of its success after all the hard work done.

However, this is not the end of assessment forever. Every six months or so there is a follow-up *surveillance* visit to check that the QMS (quality management system) is still working and continues to meet the requirements of the ISO 9001 Standard. The approach is in principle much the same as adopted during the initial assessment. The only slight difference is that less time is spent (typically one to two person days on the organization's site) because only part of the whole QMS will be covered (i.e. a sampling approach is taken). A review of internal audit, corrective action and management review etc. is usually the starting point of all such visits. As in the full assessment, Minor or even Major Non-conformities can be raised. The first *surveillance* visit is likely to focus on whether any Non-conformities raised at the assessment have been dealt with adequately. In principle, the consequence of Major Non-conformity would be revocation of the ISO certification, although in practice this is unlikely without some warning given during previous surveillance audit visit based on deficiencies found in that visit.

A few points to consider—assessment bodies vary in their practice on re-assessment frequency. Some work to a three-year cycle with a full assessment repeated at the end of this three-year period (and a new contract agreed). The argument for this policy is that after three years, it is likely that the business circumstances of the already certified organization may have changed so much that only a full-assessment is appropriate. Considering the revision cycles by the ISO organization (section 8.5) this also happens to match well to take account of any revisions to the ISO Standard itself. Normally a previous surveillance audit would address this if such a change is in the offing. Due to this, in most circumstances, assessment bodies consider continuous surveillance to be adequate rather than a full re-assessment. In this there is a point to be noted for the client organizations—they should consider re-evaluating the quotations of the external assessors. For example, "Assessor A" may be cheaper than "Assessor B" but over five years more expensive if "B" does not require full re-assessment.

8.13 ISO Consulting Services and Consultants

So far, we have discussed about the types of assessments and audit, preparing for the assessment, the assessment process. It may have made

readers aware that there are essentially three 'players' or entities involved in the implementation of the ISO based quality management system; Organization of the *user/customer*, the registrars (the external body who award the ISO Certification) and the ISO 9001:2000 Standard. Now is the time to note the forth important player; the 'consultant'. In this section, we wish to give a general overview of types of consulting services available.

8.13.1 Types of Consulting and Services

There are several types of consulting services as described below:

- **Management or Accounting Firms** who have added ISO 9000 consultancy to their list of expertise. This is usually achieved by simply hiring an expert or acquiring a consulting firm which might already have the relevant expertise.
- **Consulting Firms** which have seen the need to upgrade and expand their practice.
- **Universities and Allied Institutes** who also wish to join in the ISO 9000 market. This is usually achieved via the extended education program of the universities.
- **Individual Consultants** with varying degrees of expertise who either directly contract their skills or sub-contract to one of the above institutions.
- **Product Inspection/Testing Organizations** who have upgraded their services by including a division in charge of quality system inspection. These organizations, of which there are generally very few, usually advertise their services as facilitating the ISO 9000 implementation process (i.e. they will write and organize your quality system to suit your needs!)
- **Organizations that specialize ISO 9000 Lead Auditor or similar courses.** These organizations invariably have their head quarters in the U.K. or a commonwealth member nation.
- **Third-party Registrars** who, after much soul searching, have finally decided to join the ranks of consultants. Some registrars achieve this delicate act by creating a 'separate' consulting branch within their organization.
- **Finally**, there are consulting firms whose members usually are ex-ISO 9000 auditors/assessors. Most of these firms, offer seminars on lead auditor training and internal auditor training.

All of the above agencies will generally offer all or some of the following services or combinations thereof:

- In-house or public seminars
- Lead Auditor training courses (typically of five days duration)

- Internal Auditing courses
- Pre-assessment audit
- Third party audits (only registrars)
- Quality Manual writing
- Executive seminars for management awareness

Faced with the plethora of options, what should you do as an organization going in for ISO 9000 Certification? This leads us to the question of 'role of consultant' or 'should you hire a consultant/do you really need a consultant etc. Let us discuss this.

8.13.2 Using Consultants

Opinions on this topic vary from one extreme to another. We'll discuss both and try to present a balanced viewpoint that lies somewhere between the two extreme opinions heard. We'll also discuss what approach to take/questions to ask if you decide to hire the services of an ISO Consultant.

An extreme opinion is that the ISO Consultants are by no means essential. It is felt that if expertise is lacking in an ISO 9000 project, there are other ways of bringing it in apart from by using consultants (such as getting in your own identified persons trained on requirements of the ISO standard etc). However, the major argument in favour of consultants is that this is an effective route to buying in knowledge about the requirements of the standard and its application. The practical value of this expertise is guidance; there are real dangers of going off at a tangent and developing a system, which is either impractical, non-auditable, or simply does not meet the ISO Standard. Also, without expert guidance, the system may end up over-elaborate and far more than is truly required.

Another potential benefit of consultants is that they are outsiders and as such they can cast a fresh eye on the organizational practices and advise and recommend accordingly to bring in the quality management practices in order to meet the requirements of the Standard. However, these benefits depend on certain open-mindedness by the client organization.

Consultants can also take on a specific tasks required in building a quality (management) system—the QMS; drafting the quality manual is a good example. Here the skills and experience of a consultant may enable the job to be done more efficiently than by internal staff who generally are pre-occupied with their own high priority tasks.

However, there is also a danger in hiring the services of an ISO Consultant. It should be clearly understood (an expectations be set accordingly) that the consultants should be used primarily as facilitators and not 'doers'. This is because the ownership for the QMS should be felt by the client organization and not by the consultant. After all,

consultant is not supposed to be writing the quality manual for himself/herself. This necessitates that most of the work that goes into it has to come from internal staff. At the same time, there are 'consultants' around, who after a short briefing, will sit in a corner and draft a 'full quality system'! The result may just get the company through ISO 9000 but will otherwise be no good. Without considerable involvement by staff and commitment by the top management of the client organization, the quality management system (QMS) will not be 'owned' and consequently not effectively implemented.

8.13.3 Approach to Selecting ISO Consultants: Benefits and Disadvantages

The discussion in the previous section is not to mean that an organization should blindly accept ISO consultants. Nor does it mean that you should shun them; ISO consultants literally come 'cheaper by the dozen' as the expression goes—some really good and some very mediocre ones. In summary, there is no shortage of quality/ISO consultants to choose from having made such a decision. The choice ranges from independent freelancers to the large partnership of management consultants (in section 8.13.1, we discussed about the several types of consulting services available)

Before selecting a consultant or a consulting organization, you should ask a few questions that help you to:

- Assess the consultant's area(s) of expertise
- Determine the number organizations where he/she has provided assistance (i.e. client referral check) and
- Perform a check on a couple of references

In author's opinion, organizations should only sub-contract for certain types of ISO 9000 consulting services. There are basically two things that a hiring organization *should not* ask consultant to do:

- Writing quality manual
- Writing Quality Assurance Processes

The reasons should be obvious—in the light of discussion in the previous section. If a consultant is hired to put a QMS (quality management system) in place, then the consultant better be present on the days of the third party audit for he will likely have become the one expert with the most knowledge about "your" quality assurance system. In such cases, your chances of passing a third party audit are actually significantly reduced.

Another disadvantage of hiring a consultant to document your quality assurance system is that it usually is somewhat expensive.

Naturally, this is good for the consultant! When you hire a consultant to document your processes, including your quality manual, you must pay him/her to first understand the "ins and outs" of your industry; thus the consultant is actually "learning" at your cost!! Depending on the complexity of the business operations of your organization, this may easily take as much as a couple of weeks even if the consultant is highly experienced and has a sharp grasp to understand the subtleties and complexities of your organization's business operations and software life cycles. Besides, why would you want to pay someone to tell him/her what and how you do something, so that he/she can write it down for you? You, in turn, will have to edit what has been written down to ensure that it is correct! Would you indeed want to pay for this "service" simply because you do not have the time to do it? This is a serious question to consider while making a decision about hiring an ISO consultant. What if your management is willing to allocate adequate time or resources to ensure that your ISO implementation process will succeed?

Having discussed the pros and cons of involving ISO consultants, this section cannot be concluded without offering some positive comments regarding their role. Consultants can be most effective in a variety of roles. For example, they can:

- Offer training
- Offer recommendations based on their varied experience
- Act as a catalyst to motivate and direct people
- "Float" across departments as "neutral agent". This can be particularly helpful in situations where inter-departmental rivalries are strong. (But then also remember that many times management hires an ISO consultant to keep their gun on his/her shoulder to shoot at their own people! That is, they get their "hidden agendas" settled through the consultant. Be aware of this "game" and avoid getting into that tango).
- Act as an interface (similar to the above point)
- Guide and help monitor efforts via carefully planned periodic visits. This activity ensures that the overall implementation plan is kept 'under control'.
- Answer "difficult" questions.
- Review and edit but NOT write documentation (quality manuals etc.)
- Conduct pre-assessment audits and offer unbiased constructive opinions/suggestions.

8.13.4 Consultation Charges and Fees

Having discussed all this about the ISO Consultants, next, the readers may have a question about the fees charged by these consultants. Well,

this varies considerably depending on the mode of work and the expertise. The rates can be daily or a lump sum fee based on the tasks to be accomplished within the time schedules laid down by the client organization. To the best of author's knowledge, there are no legal guidelines on this and as a result, the consultants are free to charge fees in their own way. However while paying them some factors should be kept in mind. Some consultants may like to divide their period of professional association with the client organization in three to five broad areas and then charge differently for each phase or charge one large amount for an end-to-end involvement with their client organization. Generally, however, if an assessment body is used, their charges for pre-assessment are likely to be close to those made for the assessment itself.

As an example, we have shown below some typical phases of association with the ISO consultants (if readers recollect, section 8.13.1 indicates the types of services offered by consultants). The fees charged depend on the extent of involvement with each phase indicated below.

- "Awareness phase" (here, an awareness about quality systems is created in the client organization) based on invitation by the client
- Documentation Review
 - Identify Documents to be covered under the review
 - Gather Documents & information for review
- Assessment Planning
 - Identification of the Scope, Sponsor, Assessment and Team etc for the assessment
 - Identify "gaps" and risks based on this and establish expectations accordingly
 - Development the Assessment/Internal Audit program
 - Conduct the executive overview for the plan chalked out
- Training in Assessment/Internal Auditing Skills
 - This training is a pre-requisite for any member of the assessment team. The focus of this training is building a team during this training.
- Visit to Client organization site
 - Talk to the key members of the organization (who would be the decision makers/opinion makers and influencers)
 - Consolidate findings
 - Prepare and present draft findings
 - Prepare and deliver final findings

Prior to the phases indicated above, the organization may have already completed the following with the help from a suitable ISO consultant:

- Preparation of Draft Quality Manual (major inputs and involvement by the staff or the client organization)

- Adequacy check on the draft manual prepared
- Identify additional gaps identified, modifications to processes accordingly and revision of the Quality Manual
- Decision of the number of Internal Audit cycles as required
- Identify the third party certification agency for the client organization

The ISO consultants are expected to do handholding with the client organization until a dialogue is generated between the client organization and the third-party final auditing/assessment agency (the registrar organization). As discussed towards the end of section 8.12, client organizations should consider re-evaluation of assessment body's quotation.

8.14 ISO 9000: Some FAQs

Finally, let us look into some 'frequently asked' questions regarding ISO (though we have already dealt with some of them so far).

What does "international standardization" mean?

When the large majority of products or services in a particular business or industry sector conform to International Standards, a state of industry-wide standardization can be said to exist. This is achieved through consensus agreements between national delegations representing all the economic stakeholders concerned—suppliers, users and, often, governments. They agree on specifications and criteria to be applied consistently in the classification of materials, the manufacture of products and the provision of services. In this way, International Standards provide a reference framework, or a common technological language, between suppliers and their customers—which facilitates trade and the transfer of technology.

What benefits does international standardization bring to businesses?

For businesses, the widespread adoption of International Standards means that suppliers can base the development of their products and services on reference documents, which have broad market relevance. This, in turn, means that they are increasingly free to compete on many more markets around the world.

What benefits does international standardization bring to customers?

For customers, the worldwide compatibility of technology, which is achieved when products and services are based on International

Standards, brings them an increasingly wide choice of offers, and they also benefit from the effects of competition among suppliers.

What is 'ISO Memento'?

This was addressed in section 8.4

Is ISO planning to do something in the prominent area of 'e-Business'?

This is addressed in the next section.

8.15 e-Business and ISO

e-business (Electronic Business) is becoming a cornerstone of the world economy. Full benefits for consumers, industry and government demand a coherent set of Information and Communication Technology standards which are:

1. Open
2. Interoperable
3. Internationally accepted

To serve this aim, a Memorandum of Understanding (MoU) on electronic business in support of e-commerce has been signed by the four main organizations which develop international standards in this area: ISO, IEC, ITU, UN/ECE, with full participation from international user groups.

The MoU establishes a coordination mechanism under a unique cooperative model to produce mutually supportive standards required in business transactions (data interchange and interoperability) as well as products design and manufacturing to meet the urgent needs of both the industry and the end-users. Electronic business covers the information definition and exchange requirements within and between enterprises, including customers. Given that it provides the vital framework for e-commerce, it is intended that this MoU will support this rapidly changing and fast growing business sector.

The purpose of the MoU is to minimize the risk of divergent and competitive approaches to standardization, avoid duplication of efforts and avoid confusion amongst users. The MoU is open to other international, regional, governmental, industry and consumer organizations whose core mission involves standards setting. The MoU vision is to offer the environment in which all key international organizations can cooperate and contribute to the delivery and promotion of the evolving set of e-business standards, and maximize their contribution to global commerce.

The objective of the MoU is to encourage inter-operability by:

- Recognizing the risk of divergent or conflicting approaches to standardization
- Avoiding duplication of efforts and, therefore, confusion amongst users
- Ensuring inter-sectoral coherence

More information may be obtained from the MoU site at:

<http://www.itu.int/itu-t/e-business/mou/>

Conclusion

The aim of this chapter was to expose readers to many known as well as not-so-well-known aspects of ISO 9001. ISO as an organization is a very large body with international membership spread across countries. The revision process of ISO is an elaborate one wherein its members get the opportunity to express their opinions. This is because all strategic decisions are referred to the "ISO Members" who meet for an annual General Assembly.

With ISO 9001, organizations get a framework for their endeavour toward continual improvement. The new ISO 9001:2000 has come up with some additional requirements. It has ingeniously combined the scattered clauses of the older version into fewer but well organized newly titled clauses. This gives it a structure in line with most eminent process improvement models.

Opinions on utilizing the services of ISO consultants vary widely but both the extremes have their own justifications. The nature of consultants' involvement with client organizations can take many forms and the professional engagements depend on the ultimate objective of the client organization.

ISO has turned its attention to the upcoming area of e-Business and a MoU has been formed towards serving this end. ISO will continue to remain one of the most popular standard for process improvement and quality management system.

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SOFTWARE CMM AND OTHER PROCESS IMPROVEMENT MODELS

9.1 Overview

In previous chapters we have mentioned about the Software Engineering Institute (famously known as SEI) at Carnegie Mellon University, USA. As readers may be aware, their site address is www.sei.edu/cmu. By visiting this, readers can get details about the work carried out by SEI. In this chapter, our objective is to provide an overview of the *process improvement models* in general and particularly about the ones from the SEI. In doing this, we will be making a passing reference to the P-CMM model only for completeness. This is because P-CMM is a model for "people maturity" related practices in organizations and not about "software maturity" related practices. As a background to this, it is important to discuss the term "maturity" as it found that at times, there is some confusion about this term. This will be discussed in the context of the "Capability Maturity Model". We'll also provide some insights into what are the practices followed at "mature organizations". We'll also discuss about benefits of CMM-Based Software Process Improvement. As the ISO model also is quite popular, we'll provide a comparison of the SW CMM model with the ISO models. (Chapter 8 is devoted to discussion on the ISO 9001:2000 model). We now start with discussion of some basic terms.

9.2 The Capability Maturity Model for Software: An Overview

The Capability Maturity Model for Software (SW-CMM) developed by the Software Engineering Institute (www.sei.cmu.edu) has had a major influence on software process and quality improvement around the world. Although the SW-CMM has been widely adopted, there remain many misunderstandings about how to use it effectively for business-driven software process improvement. The SW-CMM defines a

five-level framework for how an organization matures its software process. These levels describe an evolutionary path from ad hoc, chaotic processes to mature, disciplined software processes. Table 2 shows the five levels, and the 18 *key process areas*. Figure 1 depicts the structure of the SW-CMM Framework. The five levels can be briefly described as shown in Table 1.

Table 1: Description of Software CMM Maturity Levels

Software CMM Maturity Level	Description of Software CMM Maturity Levels
1) <i>Initial</i>	The software process is characterized as ad hoc, and occasionally even chaotic. Few processes are defined, and success depends on individual effort and heroics.
2) <i>Repeatable</i>	Basic project management processes are established to track cost, schedule, and functionality. The necessary process discipline is in place to repeat earlier successes on projects with similar applications.
3) <i>Defined</i>	The software process for both management and engineering activities is documented, standardized, and integrated into a standard software process for the organization. All projects use an approved, tailored version of the organization's standard software process for developing and maintaining software.
4) <i>Managed</i>	Detailed measures of the software process and product quality are collected. Both the software process and products are quantitatively understood and controlled.
5) <i>Optimizing</i>	Continuous process improvement is enabled by quantitative feedback from the process and from piloting innovative ideas and technologies.

The key process areas are satisfied by achieving goals. Goals are described by key practices, sub-practices, and examples. The rating components of the SW-CMM are maturity levels, key process areas, and goals. The other components are informative and provide guidance on how to interpret the model. Appendix A has information on the Indian Software Organizations that have been assessed at maturity levels 4 and 5.

Except for Level 1, each maturity level is decomposed into several *key process areas* that indicate the areas an organization should focus on to improve its software process. The description of key process areas in Version 1.1 of the Software CMM is shown in Table 1. For convenience, the key process areas are internally organized by *common features*. The common features are attributes that indicate whether the implementation and institutionalization of a key process area is effective, repeatable, and lasting.

The five *common features* are *Commitment to Perform*, *Ability to Perform*, *Activities Performed*, *Measurement and Analysis*, and *Verifying*

Implementation. General practices that apply to every key process area at every maturity level are categorized by the *common features*. For example, establishing policies is a common practice in Commitment to Perform and providing training is a common practice in Ability to Perform. Each key process area is described in terms of the key practices that contribute to satisfying its goals and that are allocated to the common features. The key practices describe the specific infrastructure and activities that contribute most to the effective implementation and institutionalization of the key process area. For a full description of the SW CMM Model version 1.1, readers are directed to [2].

It is important to note that while the SW-CMM has been very influential around the world in inspiring and guiding software process improvement (SPI), it has also been misused and abused by some and not used effectively by others.

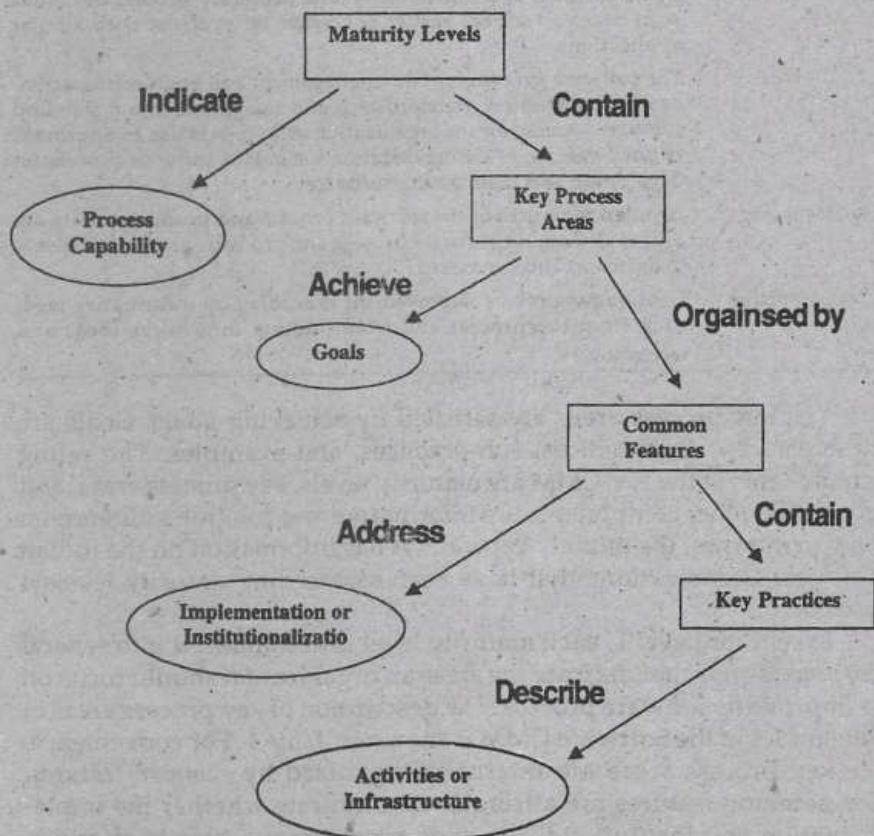


Figure 1: The CMM Structure

Table 2: The Key Process Areas in the SW-CMM version 1.1

Level	Focus	Key Process Areas
5 Optimizing	Continuous Process Improvement	• Defect Prevention • Technology Change Management • Process Change Management
4 Managed	Product and Process Quality	• Quantitative Process Management • Software Quality Management
3 Defined	Engineering Processes and Organizational Support	• Organization Process Focus • Organization Process Definition • Training Program • Integrated Software Management • Software Product Engineering • Inter-group Coordination • Peer Reviews
2 Repeatable	Project Management Processes	• Requirements Management • Software Project Planning • Software Project Tracking & Oversight • Software Subcontract Management • Software Quality Assurance • Software Configuration Management
1 Initial	Competent people and heroics	

9.3 Practices Followed at "Mature Organizations"

Having discussed, what "maturity" means in terms of "software processes", it will now be interesting to understand the practices followed in mature organizations. Over the last few years the Software Engineering Institute has participated in several workshops and site visits with maturity level 4 and 5 software organizations. A detailed discussion on this is provided in Appendix A. A net summary based on SEI's survey shows that the following factors/approaches are found to be present in "high maturity organizations":

- Quality Culture and Process Orientation
- Commitment to TQM
- ISO 9001 certification
- Effective Addressing of people issues
- Data-driven decision making
- Customer education when working with lower maturity customers

9.4 CMM and ISO (A Comparative Analysis)

"How does ISO compare with the CMM?"—this is one of the most frequently asked questions. People wish to know which model should an organization choose. Another question that people often come up with

is "if an organization has the ISO certification, then at what (SW) CMM maturity level does this correspond to. In this section, we'll take up a discussion towards this. We'll provide a comparison analysis but we'll not make any prescriptive statement; we prefer to leave it to the reader to draw his or her own conclusions based on the analysis provided here.

9.4.1 CMM and ISO 9001

The Capability Maturity Model for Software, developed by the Software Engineering Institute, and the ISO 9000 series of standards, developed by the International Organization for Standardization, have the common concern of quality and process management. The two are driven by similar issues and are intuitively correlated, but they differ in their underlying philosophies: ISO 9001, the standard in the 9000 series that pertains to software development and maintenance, identifies the minimal requirements for a quality system, while the CMM underlines the need for continuous process improvement. This statement is somewhat subjective, of course (especially, as we'll see in the Chapter on ISO, that the new revised year 2000 release of ISO talks about "Continual Improvement"). Some members of the international standards community maintain that if you read ISO 9001 with insight, it does address continuous process improvement. Corrective action, for example, can be construed as continuous improvement. Nonetheless, according to Marc Paul of the SEI [7] the CMM tends to address the issue of continuous process improvement more explicitly than ISO 9001. For a complete discussion on this topic, readers should refer [7] wherein a complete comparative analysis is provided between the CMM model and the ISO standard by taking up each clause of the ISO. Table 3 provides the summary of Marc Paulk's mapping between ISO 9001 and the CMM. However, this analysis has been extended by adding the additional column to show the mapping of SW CMM to the new ISO 9001:2000.

Table 3: Summary Mapping Between ISO 9001 And The CMM

ISO 9001:1994 Clause	ISO 9001:2000 Clause	Strong Relationship with CMM	Judgmental Relationship with CMM
4.1: Management Responsibility	4.1: General Requirement	Commitment to perform	• Ability to perform
	4.2.1: General		• Verifying
	5.1: Management Commitment	Software project planning	Implementation
	5.3: Quality Policy	Software project tracking and oversight	• Software quality management
	5.4.1: Quality Objectives	Software quality assurance	

ISO 9001:1994 Clause	ISO 9001:2000 Clause	Strong Relationship with CMM	Judgmental Relationship with CMM
4.2: Quality system	4.1: Quality Management Systems: General Requirement	Verifying implementation Software project planning	• Organization process definition
	4.2.1: Documentation Requirements: General	Software quality assurance	
	4.2.2: Quality Manual	Software product engineering	
4.3: Contract review	7.2.3: Customer Communication	Requirements management Software project planning	• Software subcontract management
4.4: Design control	7.3.1: Design and Development Planning	Software project planning	• Software quality management
	7.3.2: Design and Development Inputs	Software project tracking and oversight Software configuration management	
	7.3.3: Design and Development Outputs	Software product engineering	
	7.3.4: Design and Development Review		
	7.3.5: Design and Development Verification		
	7.3.6: Design and Development Validation		
	7.3.7: Control of Design and Development Changes		
4.5: Document and Data control	4.2.3: Control of Documents	Software Configuration Management Software Product Engineering	
4.6: Purchasing	7.4.1: Purchasing Process	Software subcontract management	
	7.4.2: Purchasing Information		
	7.4.3: Verification of Purchased Product		

<i>ISO 9001:1994 Clause</i>	<i>ISO 9001:2000 Clause</i>	<i>Strong Relationship with CMM</i>	<i>Judgmental Relationship with CMM</i>
4.7: Control of Customer-Supplied Product	7.5.4: Customer Property	—	• Software subcontract management
4.8: Product Identification and Traceability	7.5.3: Identification and Traceability	Software configuration management Software product engineering	—
4.9: Process control	6.3: Infrastructure	Software project planning	• Quantitative process management
	6.4: Work Environment	Software quality assurance	• Technology change management
	7.5.1: Control of Production and Service Environment	Software product engineering	
	7.5.2: Validation of Processes for Production and Service Provision		
4.10: Inspection and Testing	7.5.3: Identification and Traceability	Software product engineering	
	8.2.4: Monitoring and Measurement of Product	Peer reviews	—
4.11: Control of Inspection, Measuring, and Test equipment	7.6: Control of Monitoring and Measuring Devices	Software product engineering	—
4.12: Inspection and Test Status	—	Software configuration management Software product engineering	—
4.13: Control of Nonconforming Product	—	Software configuration management Software product engineering	—
4.14: Corrective and Preventive Action	8.5.2: Corrective Action	Software quality assurance	• Defect prevention
	8.5.3: Preventive Action	Software configuration management	
4.15: Handling, Storage, Packaging, Preservation, and Delivery	7.5.5: Preservation of Product	—	• Software configuration management • Software product engineering

ISO 9001:1994 Clause	ISO 9001:2000 Clause	Strong Relationship with CMM	Judgmental Relationship with CMM
4.16: Control of Quality Records	4.2.4: Control of Records	Software configuration management Software product engineering Peer reviews	-----
4.17: Internal Quality audits	8.2.2: Internal Audit	Verifying implementation Software quality assurance	-----
4.18: Training	6.2.2: Competence, Awareness and Training	Ability to perform Training program	-----
4.19: Servicing	-----	-----	-----
4.20: Statistical Techniques	8.2.4: Monitoring and Measurement of Product 8.4: Analysis of Data		• Organization process definition • Quantitative process management • Software quality management

9.4.2 CMM and 15504

This section provides an overview of the Software CMM and ISO/IEC 15504, discusses the similarities and differences between them, and speculates how they may influence each other as they both continue to evolve. ISO/IEC 15504 is a suite of standards for software process assessment currently under development as an international standard. ISO/IEC 15504 has been published as a type 2 technical report, which is a stage in the development of a standard. Of the nine parts to ISO/IEC 15504, the parts directly relevant to the Software CMM are ISO/IEC 15504-2, the reference model, and ISO/IEC 15504-5, which provide an example model. As an international standard, ISO/IEC 15504 can be expected to affect the continuing evolution of CMM-related products. ISO/IEC 15504 is being developed by working group 10 (WG10) under the software engineering subcommittee (SC7) of the information technology joint committee (JTC1) established by the International Organization for Standardization (ISO) and the International Electro-technical Commission (IEC). For a detailed discussion on this, reader should refer to [8].